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Designing for Life: A Socioeconomic View of Digital Learning Preferences in Cybersecurity, with Emphasis on Older Adults

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Abstract

As digital literacy becomes central to cybercrime prevention, we examine how adults of different ages engage with online learning, moving beyond age alone to consider additional drivers of preference. We analyzed a nationally representative U.S. adult sample (N = 1113; Nov 2024). Ordinal logistic regressions assessed associations between preferences for cybersecurity education and age, education, income, subjective well-being (SWB), and high-speed internet access. Interaction terms (e.g., age × internet access) were tested but not retained. Preferences declined with age across most tools, with the sharpest drop being for highly interactive or novel formats (VR/AR, gamification). Actor-based, non-interactive videos showed no age advantage. Education displayed selective positive links, especially for interactive features, while income was largely unrelated. SWB was a broadly enabling correlate, often with nonlinear patterns, and reliable high-speed internet was consistently aligned with stronger preferences. Overall, the model fit was moderate. Effective cybersecurity education should not rely on age-based assumptions. Designing offerings that emphasize clear purpose and ease of use, pair reliable broadband with skills supports, and account for learners' well-being can improve engagement and potential scam resilience across age groups.

Keywords: cybercrime prevention; online learning; subjective well-being; broadband access; gamified education; older adults; cybersecurity training

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1. Introduction and Scope of Study

Cybersecurity education and awareness programs remain essential tools for engaging people of all ages with the realities of contemporary threats. Recent reviews show that user-focused, "target-hardening" interventions—such as simulated phishing, scenario-based modules, and campaign toolkits—systematically cover current and emerging risks and can improve detection/reporting when designed and evaluated well; at the same time, they caution that not all programs change behavior equally and that content and delivery matter [1].

Age shapes how people learn and what learning formats they prefer. Older learners learn better with slower pacing, clear feedback, and multiple types of support. Meta-analytical and qualitative evidence also indicate that age moderates the motivation-to-learning transfer and that older adults favor hands-on, step-by-step materials for technology training [2].

Focusing on cybersecurity even in later life is warranted because older adults bear disproportionate financial harm from online crime. In 2024, people aged 60+ filed > 147,000 complaints and reported almost $4.9 billion in losses (more than any other age group), with an average loss of >$83,000 and 7500+ victims each losing $100,000+, according to the Federal Bureau of Investigation [3]. Consistent with this, the Federal Trade Commission's most recent report finds that while older adults are less likely to report any monetary loss than those aged 18–59, those who do report losses have higher median losses ($650 for 60+ vs. $450 for 18–59; $1450 for age 80+) and are far more likely to lose money to particular frauds, controlling for population size [4].

Although most cybersecurity education programs focus on younger adults [5], we must also purposefully offer cybersecurity training that is tailored for all age groups. Specifically, older adults are markedly under-represented in cybersecurity education, and if they are included, they are often seen as a homogeneous group [5]. Additionally, we still know little about which social and demographic factors (e.g., education, income, gender, digital experience) shape adults' cybersecurity learning preferences. Scoping and systematic reviews describe heterogeneous findings across settings and measures, and even the instruments used to assess "digital literacy" (including information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving [6]) vary widely—factors that help explain the mixed results reported to date [7]. This study addresses this gap by asking US adults on a national sample, that is representative of age, which online cybersecurity learning tools they prefer. Building on these findings, we provide actionable, age-responsive strategies for designing cybersecurity education programs that effectively engage learners across the adult lifespan. We specifically outline solutions for designing cybersecurity education that equitably incorporates older age groups, a frequently overlooked population in current programming.

2. Literature Review and Theoretical Framework

Building on three complementary frameworks, we expect a negative age gradient in preferences for online learning. Andragogy [8] holds that adults learn best when instruction is purpose-driven, self-directed, and experience-anchored; when a format obscures why it matters or constrains autonomy, adults disengage. Socioemotional selectivity theory (SST; Carstensen et al. [9]) adds that, with age, perceived time horizons shorten and learners increasingly prioritize meaning and immediate utility over novelty or exploration. The technology acceptance model (TAM; [10]) predicts adoption only when tools are perceived as useful and easy to use—thresholds that typically become stricter with age as costs of effort and error rise. Each of the selected three frameworks explains a different, age-relevant barrier in online learning: andragogy speaks to adults' need for purposeful, autonomy-supportive instruction; SST explains why older adults increasingly prioritize meaningful, low-effort learning; and TAM captures how usability and perceived usefulness become stricter thresholds with age. Together, these theories show that aging increases learners' focus on usefulness, clarity, and low cognitive effort.

Empirical evidence is consistent with this account. In laboratory and field studies, goal-directed selectivity increases with aging, choosing fewer but more personally meaningful learning opportunities [11,12]. A meta-analysis of adult training confirms that age moderates transfer: motivation translates less into complex performance for older learners, especially under heavy strategy or multitask demands [13]. Cognitive and neural work points to mechanisms that raise the cost of effortful learning with age—dopaminergic changes that blunt reward-based learning and executive-control limits that tax multi-step, rule-based updating [14,15]. Studies on technology training further show preferences for slower pacing, stepwise guidance, and stable interfaces, rather than exploratory, multi-

window interaction [2,16,17]. Taken together, these findings imply that when tools are confusing, unfamiliar, or not clearly useful, older adults are less likely to accept them, leading to a lower preference for online learning overall. Hence, we expect the following:

H1. *As age increases, preference for online learning mechanisms, in general, will decline.*

These same principles imply a steeper age penalty for highly interactive, gamified formats such as VR/AR. Interactivity introduces novel controls, rapid decision cycles, and sensorimotor demands that elevate the extraneous cognitive load, lowering perceived ease of use and sometimes perceived control. Andragogy's self-directed learner can react negatively when a platform feels like the tool is "driving" the session; SST further tilts older learners toward efficient, goal-focused engagement, rather than open-ended exploration [7,18]. Consistent with this, technology-training studies report preferences among older age groups for slower pacing, stepwise guidance, and stable interfaces over fast, exploratory interaction [2,16,17]. Meta-analysis likewise indicates that age dampens gains when training is complex or highly interactive [13]. Thus, we hypothesize that with an absence of strong scaffolding and immediately visible payoff,

H1a. *Interest in VR/AR-style interactivity should be lower among older than younger age groups.*

By contrast, less interactive but concrete, story-anchored formats fit older adults' selection criteria. Actor-led videos showing real-life case scenarios leverage andragogy's experience-based learning (new content mapped onto recognizable situations) while preserving autonomy through self-paced viewing and replay [16,19]. They satisfy SST's motive for personally meaningful, problem-focused content [18] and score high on TAM's perceived usefulness with minimal interface burden [10]. Field evidence underscores the point: during COVID-19, brief, purpose-framed training more than doubled older age groups' telehealth adoption, illustrating that clear relevance—not novelty—drives uptake [20]. Community- and service-anchored learning similarly taps generativity motives and sustains engagement without heavy interactivity [21], and targeted memory/strategy supports further improve performance in story-rich contexts [22]. In addition, theater-based education works well for older adults who favor personal stories and in-person formats [23], as opposed to younger age groups, who are more receptive to digital, multi-channel, and hybrid modes of education [24]. We therefore expect the following:

H1b. *Greater interest among older adults in actor-based, real-life video scenarios than in younger age groups, even as overall enthusiasm for online mechanisms declines with age.*

Additional demographic factors beyond age also shape learning preferences. Digital divide theory emphasizes that unequal access to digital technologies restricts older adults' learning opportunities and participation [25]. Crucially, this divide is multilayered—encompassing motivational, material, skills, and ICT usage access—and these dimensions are strongly patterned by income and education.

Evidence consistently shows that higher-income older adults have greater access to devices and broadband and, consequently, higher levels of internet use and digital engagement. Choi and DiNitto report that older adults with more education and income are significantly more likely to use the internet, underscoring how financial resources facilitate technology adoption [26]. Complementary findings show that digital literacy training designed for low-income older adults improves effective technology use, highlighting the role of socioeconomic status (SES) in shaping access to skill-building opportunities [27]. SES also influences how older adults use technology for social participation: engagement

with online activities—including social media and telehealth—is higher among those with more resources [28,29]. In addition, during the COVID-19 pandemic, older adults with higher incomes were more likely to adopt telehealth, due to stronger digital readiness [30].

Digital literacy further mediates these SES effects. Income-advantaged older adults have more opportunities to develop digital competence, which increases comfort, proficiency, and the likelihood of engaging in online activities [31,32]. Conversely, those with fewer financial resources face barriers that impair digital engagement and can negatively affect mental health and social connectedness [33,34]. Community-level interventions such as subsidized broadband and targeted training programs can help mitigate these disparities, though limited access to devices and skills continues to reinforce isolation for the lowest-income groups [30,35]. Thus, we hypothesize the following:

H2. *Income level will positively correlate with openness to various online methods.*

Education shows similar patterns. Adults with higher educational attainment are more likely to participate in online activities—from social networking to seeking health information—due to higher digital literacy and greater technological self-efficacy [32]. This pattern appears across international contexts, where lower educational attainment predicts reduced use of digital health services [36]. Tailored training programs that improve attitudes toward technology can enhance internet engagement among older adults [37]. In addition, socioeconomic background shapes not only access and ability but also the psychological resources needed for sustained digital participation [26]. Educational disparities extend to health behavior as well: those with more education are more likely to use digital tools for managing health information [38], reflecting broader differences in e-health literacy [39]. Participation in offline cultural activities, which correlates with education, also supports more confident online behavior [40]. Across the literature, education consistently emerges as a key determinant of both the quantity and quality of older adults' digital participation, with inclusive training shown to strengthen autonomy and digital engagement [41]. Hence, we hypothesize the following:

H3. *Education level will positively correlate with openness to various online methods.*

Taken together, education and income do not only shape older adults' access, skills, and confidence for online learning; they also determine how much value—social connection, support, and purpose—they can realistically extract from it. In turn, when online engagement reliably enhances subjective well-being (SWB) through community and support, it becomes more attractive. This explains why preferences for specific formats vary with both socioeconomic resources and expected SWB gains. Accordingly, older adults' subjective well-being is not solely determined by traditional parameters, such as health and social interactions, but is also significantly influenced by their engagement with technology. Studies suggest that online learning can foster a sense of community among adults, which is critical for enhancing their SWB. For instance, social support derived from online platforms can positively affect emotional well-being, promoting feelings of belonging and satisfaction [42]. Specifically, Lin et al. highlight the role of community engagement, where the nexus of social capital and intergenerational support further enhances the well-being of older adults [43].

The psychological dimensions of online education have been explored, with findings underscoring that such platforms can alleviate feelings of isolation while improving motivation and self-efficacy [44]. Research indicates that when adults engage enthusiastically in online learning, they report lower levels of anxiety and depression, further enhancing their SWB [45]. Self-efficacy, defined as an individual's belief in their ability to execute

the behaviors necessary to achieve specific performance outcomes, positively correlates with emotional regulation and resilience among the elderly [46]. An engaged lifestyle not only enhances cognitive abilities but also fosters increased social interactions and community involvement, which are essential elements in supporting the well-being of older populations [47]. Therefore, we assume the following:

H4. *Subjective well-being will be associated with greater preferences for online learning methods.*

Finally, because most contemporary e-learning relies on streaming video, synchronous discussion, and interactive content, reliable broadband is a prerequisite for meaningful participation. Research posits that households without a reliable, robust connection cannot fully take part in virtual courses and trainings (e.g., video drops, inability to join live sessions), and broadband availability/adoption is directly tied to participation in online schooling and other digital learning activities [48]. Minimum bandwidth studies likewise show that common platforms require multi-Mbps connections, well beyond an unreliable or mobile-only service, underscoring why reliability and speed matter for online course access and completion [49]. Research further documents that where home broadband is present and dependable, adults report higher levels of internet use across work, health, and education domains. In contrast, where it is absent or unstable, engagement falls off, especially for older and lower-income households [50]. In our models, we therefore expect the following:

H5. *Reliable, high speed internet access will be associated with stronger preference for online learning methods.*

3. Sample and Methods

This study investigated the online learning preferences of different age groups in the United States, based on a sample (N = 1113) derived from a US national survey, representative by age, in November 2024. The research was reviewed by Virginia Tech's Institutional Review Board #24-1207. Data collection was completed using Qualtrics, through CloudResearch, an online survey recruitment and fielding agency. Professional middleman services are non-probability sampling agents that recruit participants according to population representation criteria given by researchers or businesses and field online surveys in return for a moderate participant compensation. Social research frequently applies these services (e.g., Reichelmann and Costello [51]; Dearden et al. [52]; Hawdon et al. [53]; Kemp et al. [54]) and demonstrates their reliability in representing population demographics and other attributes [55–57]. Although they do track participant IP addresses in order to provide minimal compensation, they do not share participants' personal identification data with the researchers. Hence, participation in our survey was voluntary and anonymous. The average time for participants to complete the survey was approximately 6.94 min. The initial data collection yielded 1097 participants, but 14 participants were dropped from the sample for speeding (completed the survey under 180 s), and an additional 96 participants were dropped for not passing the attention check in the middle of the approximately eight minute survey. The final dataset includes 1003 participants. In our sample, females were also slightly overrepresented (65.4%). Participants were given the definition of high-speed internet as follows: "In 2024, the Federal Communications Commission defines broadband internet as having a minimum download speed of 25 Mbps and an upload speed of 3 Mbps." Our sample participants were slightly more educated than the US average, and 65+ populations were slightly overrepresented: both are features of prolific non-probability samples [57]. In our sample, 91.8% reported high-speed internet, which is slightly lower than the official statistics (95% of US homes and small businesses had

high speed internet [58]); however, 20.2% of people with high-speed internet in our sample reported having sometimes slow or unreliable connections. The demographic composition, compared to US population statistics, is presented in Table 1.

Table 1. Sample demographics compared to US Census.

Sample		US Census
Sex		
Female (18+)	656 (65.4%)	134.3 M (50.9%) *
Male (18+)	347 (34.6%)	129.2 M (49.0%) *
Total (18+)	1003 (100.0%)	263.5 M (100.0%) *
Age group (Recoded)		
18–44	440 (43.9%)	116.9 M (43.8%) **
45–64	302 (30.1%)	88.6 M (33.2%) **
65 and above	261 (26.0%)	61.2 M (22.9%) **
Total	1003 (100.0%)	266.7 M (100.0%) **
Education		
Less than high school	35 (3.5%)	24.3 M (9.4%) ***
High school diploma or equivalent	332 (33.5%)	74.6 M (28.9%) ***
Some college	232 (23.4%)	42.6 M (16.5%) ***
Bachelor's degree	163 (16.5%)	57.3 M (22.2%) ***
Associate degree	131 (13.2%)	25.6 M (9.9%) ***
Master's degree	75 (7.6%)	25.6 M (9.9%) ***
Doctoral or professional degree	23 (2.3%)	8.5 M (3.3%) ***
Total	991 (100.0%)	258.5 M (100.0%) ***
Income (Recoded)		
Less than $25,000	235 (24.6%)	128.2 M (13.5%) +
$25,000–$50,000	309 (32.4%)	22.5 M (16.7%) +
$50,001–$75,000	192 (20.1%)	20.4 M (15.1%) +
More than $75,001	219 (22.9%)	73.9 M (54.8%) +
Total	955 (100.0%)	134.8 M (100.0%) +
Subjective well-being: Compared to others in the US, how well do you feel yourself?		
A lot better off	101 (10.2%)	n/a
A little better off	226 (22.8%)	n/a
Just the same	336 (33.9%)	n/a
A little worse off	236 (23.8%)	n/a
A lot worse off	92 (9.3%)	n/a
Total	991 (100.0%)	n/a
High-speed internet		
High-speed internet, reliable	731 (73.8%)	n/a
High-speed internet but sometimes slow and unreliable	198 (19.9%)	n/a
No high-speed internet but reliable	45 (4.5%)	n/a
No high-speed internet and not reliable	6 (0.6%)	n/a
No consistent access to internet	11 (1.1%)	n/a
Total	991 (100.0%)	n/a

US Official Stats sources: * Census Population Estimates Program (PEP), Vintage 2024—the resident population as of 1 July 2024; ** ACS 2024 1 year S0101 table (survey estimate); *** U.S. Census Bureau, CPS Detailed Tables, 2024, adults 18 and over; + CPS ASEC, calendar year 2024, unit = households.

3.1. Analytic Plan

3.1.1. Dependent Variables

In an online survey, participants were asked about their preferences for online learning on cybersecurity issues. First, we asked participants a multiple-answer question about which format they feel is most engaging for online learning. Possible answers included the following: video lectures, interactive quizzes, case studies, role-playing scenarios, live group discussions, and individual study resources (e.g., articles, PDFs). Next, we modeled learning preferences as ordered categorical outcomes derived from single-item questions Q2–Q12. All items were coded so that higher values indicate greater endorsement of the stated preference (coding in parentheses). The list of Q2–12 is as follows: Importance of interactivity (Q2): 1 = Not at all important . . . 5 = Extremely important. Preferred frequency of interactive elements (Q3): recoded so that higher values indicate more frequent interaction (1 = Minimal interaction . . . 5 = Every ~5 min). Interest in actor-based scam videos (Q4): 1 = Not at all interested . . . 5 = Very interested. Perceived learning enhancement from VR (Q5) and from AR (Q6): 1 = Not at all . . . 5 = Significantly. Interest in exploring realistic VR/AR scenarios (Q7): 1 = Not at all interested . . . 5 = Very interested. Preferred interaction level in VR/AR (Q8): 1 = No interaction, 2 = Low/medium, 3 = High interaction (higher = more interaction). Perceived value of in-person workshops (Q9): 1 = Not at all valuable . . . 5 = Extremely valuable. Importance of gamification (Q10): 1 = Not at all important . . . 5 = Extremely important. Preferred level of gamification (Q11): 1 = Low . . . 3 = High (higher = more gamification). Importance of age-tailored platforms (Q12): 1 = Not at all important . . . 5 = Extremely important.

3.1.2. Independent Variables

Our predictors were selected a priori to capture life-course position, structural resources, and subjective standing—domains that plausibly shape preferences for (and access to) online learning. Age was analyzed in three ordered groups (18–44, 45–64, 65+), reflecting the study's design and the oversampling of older adults; in the models, we treated this recode as an ordered factor and tested linear/quadratic trends across groups to summarize age-graded patterns efficiently. High-speed internet access was measured with a five-category indicator anchored in the definition shown to respondents—"In 2024, the Federal Communications Commission defines broadband internet as having a minimum download speed of 25 Mbps and an upload speed of 3 Mbps"—and recorded as follows: reliable high-speed internet; high-speed but sometimes slow/unreliable; no high-speed but reliable; no high-speed and not reliable; and no consistent access. For analysis, we treated this variable as ordered, so that higher values reflect greater and more reliable connectivity. Education level was captured with seven categories, ranging from less than high school through to doctoral/professional degree; we modeled it as an ordered factor to test monotonic and curvature components via orthogonal polynomials, consistent with the theory that educational attainment proxies both skill and cultural capital relevant to digital learning. Household income was measured in four brackets (less than $25,000; $25,000–$50,000; $50,001–$75,000; more than $75,001) and likewise entered as an ordered factor; this provides a coarse gradient of material resources that may condition both bandwidth and device quality, as well as tolerance for time and opportunity costs of learning online. Finally, subjective well-being (SWB) was elicited with a five-level comparative item ("Compared to others in the US, how well do you feel yourself?": a lot worse off → a lot better off). We treated SWB as an ordered indicator of perceived standing and psychosocial reserve that could shape motivational readiness for novel learning formats.

We proceed in four steps, aligned with the questionnaire's measurement and our hypotheses. For Q2–Q12 (ordinal), we compute polychoric correlations and ordinal α

to summarize internal consistency patterns and justify the optional composites above. For Q1 (multi-select), we report item prevalences only; it is excluded from inferential models due to low internal consistency. Each dependent variable (Q2–Q12) is analyzed via ordinal logistic regression (cumulative logit/proportional odds): age group (18–44, 45–64, 65+; entered with orthogonal polynomial contrasts to test linear/quadratic trends across age); high-speed internet access (ordered five-level self-report; polynomial contrasts); education (seven levels; polynomial contrasts); income (four levels; polynomial contrasts); and subjective well-being (SWB) (five levels; polynomial contrasts). We report odds ratios (OR) with 95% CIs, AIC, and Nagelkerke's R^2. Given multiple outcomes, we present both raw *p*-values and Benjamini–Hochberg FDR adjustments in the supplementary tables; the main text emphasizes effect sizes and predicted probabilities.

For each outcome, we present marginal predicted probabilities by age group, holding other covariates at observed distributions (marginal standardization). Planned contrasts address the hypotheses: (a) linear age trends for interactivity, VR/AR, and gamification (H1/H1a/H1b), (b) positive gradients for education (H3) and income (H2), and (c) positive associations for SWB (H4) and internet access (H5). As a sensitivity analysis, we will also estimate brief composites where theory and the polychoric matrix suggest unidimensionality: (a) VR/AR affinity (Q5–Q7), (b) interactivity orientation (Q2–Q3 and Q8, standardized then averaged), and (c) gamification orientation (Q10–Q11). Composites will use ordinal-reliability estimates (ordinal α) and will not replace the preregistered single-item models.

4. Results

The structural validity of the survey was measured by the strength of correlations between individual items. Spearman's correlation coefficients were calculated for Question 1 (binary; online learning format) and polychoric correlation coefficients were calculated for Questions 2–12 (ordinal scale; importance and willingness of learning through different learning formats and topics). For Question 1, the overall Cronbach's alpha was relatively low (0.367); hence, we omitted this question from further analysis. For Question 2–12, the Cronbach's alphas were 0.868 overall, indicating the reliability of the questions. The correlation matrix is presented in Table 2.

Table 2. Polychoric correlation coefficients for Q2–12.

Polychoric Correlation	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
Q2	1.000										
Q3	0.460	1.000									
Q4	0.414	0.282	1.000								
Q5	0.416	0.274	0.465	1.000							
Q6	0.389	0.255	0.482	0.830	1.000						
Q7	0.408	0.263	0.539	0.782	0.774	1.000					
Q8	0.439	0.297	0.308	0.538	0.526	0.606	1.000				
Q9	0.496	0.308	0.508	0.579	0.552	0.587	0.534	1.000			
Q10	0.544	0.314	0.392	0.502	0.521	0.545	0.599	0.581	1.000		
Q11	0.515	0.360	0.288	0.387	0.408	0.432	0.624	0.507	0.699	1.000	
Q12	0.361	0.159	0.315	0.299	0.311	0.320	0.301	0.449	0.472	0.390	1.000

In order to test our hypotheses, we ran ordinal logistic regressions, including the three age groups we created before and the dependent variables, which can be seen in Table 3. This analytical choice was based on the question formats: Q2–12 were single-choice questions measured on ordinal scales (Table 3). In the next step, we aimed to determine

which additional demographic variables—beyond age—may influence online learning preferences. By modeling these associations, we sought to uncover a broader set of factors that shape engagement with different forms of online education.

Table 3. Ordinal logistic regressions on age groups and online learning preferences.

Ordinal Logistic Regressions	n	AIC	Nagelkerke's R^2
2. Importance of interactive features	999	2799.062	0.117
3. Frequency of interactive elements	1003	2908.745	0.037
4. Scam scenarios with human actors	1003	2784.876	0.004
5. VR could enhance learning	998	2926.979	0.033
6. AR could enhance learning	957	2780.281	0.043
7. Realistic VR/AR scenarios	947	2858.610	0.057
8. Level of interaction	769	1564.133	0.160
9. In-person workshops	995	3004.276	0.049
10. Gamified elements	875	2576.310	0.171
11. Level of gamification	746	1441.729	0.189
12. Online platform tailored to age groups	981	2906.771	0.018

When only the age variable was included in the model (Table 3), we did find significant values, but the explanatory power of the model was low, and the AIC indicator also suggested that the model fit was suboptimal. Therefore, we included additional explanatory variables in the analysis. As can be seen in Table 4, although the explanatory power of the models was low, as indicated by Tjur's R^2 or Nagelkerke's R^2 values, all were closer to zero than one. When the demographic variables were added to the regression models, the associations between age groups and online learning preferences remained statistically significant. The explanatory power of the models remained low, although it increased slightly (Table 4).

Table 4. Q2–12: Ordinal logistic regressions.

	Coefficients:				AIC	Nagelkerke's R^2
	Estimate	Std. Error	z Value	Pr (>|z|)		
Q2: Importance of interactive features						
Age_group.L ***	−1.091738	0.112812	−9.677	<2e−16		
Age_group.Q	−0.138214	0.110592	−1.250	0.211387		
HS_Internet.L ***	1.858883	0.485231	3.831	0.000128		
HS_Internet.Q	−0.379421	0.426828	−0.889	0.374040		
HS_Internet.C	−0.626268	0.578655	−1.082	0.279127		
HS_Internet ^4	0.453530	0.463741	0.978	0.328084		
Education.L *	0.856151	0.358928	2.385	0.017065		
Education.Q	0.076407	0.331833	0.230	0.817891		
Education.C *	0.576377	0.268683	2.145	0.031937		
Education ^4	0.244606	0.214375	1.141	0.253862	2582.894	0.342
Education ^5	−0.002485	0.162807	−0.015	0.987823		
Education ^6	−0.207534	0.155753	−1.332	0.182711		
SWB.L **	0.537480	0.204399	2.630	0.008549		
SWB.Q ***	0.671387	0.173876	3.861	0.000113		
SWB.C ***	0.483463	0.145319	3.327	0.000878		
SWB ^4	−0.024750	0.117001	−0.212	0.832468		
income2.L	0.273970	0.141483	1.936	0.052817		
Income.Q	−0.086563	0.125559	−0.689	0.490557		
Income.C	0.086114	0.121512	0.709	0.478519		

Table 4. *Cont.*

	Coefficients:				AIC	Nagelkerke's R^2
	Estimate	**Std. Error**	**z Value**	**Pr (>|z|)**		
Q3: Frequency of interactive elements						
Age_group.L ***	−0.65023	0.11177	−5.817	5.98e−09		
Age_group.Q	−0.16851	0.10892	−1.547	0.12184		
HS_Internet.L	0.85078	0.47469	1.792	0.07309		
HS_Internet.Q	−0.08479	0.42523	−0.199	0.84196		
HS_Internet.C	−0.18438	0.52313	−0.352	0.72449		
HS_Internet ^4	−0.03567	0.42343	−0.084	0.93286		
Education.L **	1.12835	0.34535	3.267	0.00109		
Education.Q	0.12697	0.32151	0.395	0.69291		
Education.C *	0.58052	0.25958	2.236	0.02532		
Education ^4	0.14954	0.20606	0.726	0.46800	2696.149	0.273
Education ^5	0.01800	0.15844	0.114	0.90956		
Education ^6	0.15507	0.15254	1.017	0.30936		
SWB.L	−0.11912	0.19094	−0.624	0.53272		
SWB.Q	0.14676	0.16277	0.902	0.36726		
SWB.C	0.22251	0.14001	1.589	0.11199		
SWB ^4	−0.15490	0.11770	−1.316	0.18815		
Income.L	0.12879	0.14230	0.905	0.36544		
Income.Q	−0.02290	0.12630	−0.181	0.85614		
Income.C	−0.05976	0.11940	−0.501	0.61670		
Q4: Videos of scam scenarios with human actors						
Age_group.L	−0.072948	0.108839	−0.670	0.502706		
Age_group.Q	−0.134849	0.110166	−1.224	0.220935		
HS_Internet.L ***	1.676836	0.459227	3.651	0.000261		
HS_Internet.Q	0.367321	0.410676	0.894	0.371092		
HS_Internet.C	−1.063442	0.565360	−1.881	0.059972		
HS_Internet ^4	0.508349	0.453515	1.121	0.262327		
Education.L	0.403409	0.356608	1.131	0.257954		
Education.Q	−0.056510	0.333582	−0.169	0.865478		
Education.C	0.221237	0.268367	0.824	0.409722		
Education ^4	0.199304	0.213534	0.933	0.350636	2559.210	0.260
Education ^5	0.061498	0.161717	0.380	0.703738		
Education ^6	0.052098	0.153458	0.339	0.734239		
SWB.L	0.049386	0.204110	0.242	0.808814		
SWB.Q ***	0.611153	0.173084	3.531	0.000414		
SWB.C	0.182677	0.143178	1.276	0.201999		
SWB ^4	−0.090096	0.117103	−0.769	0.441669		
Income.L	0.155653	0.141384	1.101	0.270930		
Income.Q	−0.006692	0.125848	−0.053	0.957595		
Income.C	0.114193	0.121697	0.938	0.348069		
Q5: VR could enhance learning						
Age_group.L ***	−0.52509	0.11009	−4.770	1.85e−06		
Age_group.Q	−0.07957	0.10841	−0.734	0.462965		
HS_Internet.L ***	1.96295	0.48786	4.024	5.73e−05		
HS_Internet.Q	−0.84535	0.43322	−1.951	0.051016		
HS_Internet.C	−0.62417	0.52945	−1.179	0.238443	2701.451	0.282
HS_Internet ^4	0.70226	0.42611	1.648	0.099344		
Education.L	−0.10678	0.35017	−0.305	0.760420		
Education.Q	−0.20052	0.32887	−0.610	0.542040		

Table 4. *Cont.*

	Coefficients:				AIC	Nagelkerke's R^2
	Estimate	**Std. Error**	**z Value**	**Pr (>|z|)**		
Education.C	−0.20188	0.26443	−0.763	0.445180		
Education ^4	−0.20467	0.21020	−0.974	0.330202		
Education ^5	−0.12286	0.16056	−0.765	0.444135		
Education ^6	−0.17613	0.15350	−1.147	0.251195		
SWB.L ***	0.73508	0.20062	3.664	0.000248		
SWB.Q	0.32974	0.16920	1.949	0.051320	2701.451	0.282
SWB.C *	0.37072	0.14421	2.571	0.010147		
SWB ^4	−0.13008	0.11595	−1.122	0.261898		
Income.L *	0.34055	0.14066	2.421	0.015478		
Income.Q	0.06184	0.12542	0.493	0.621940		
Income.C	0.06965	0.12077	0.577	0.564118		
Q6: AR could enhance learning						
Age_group.L ***	−0.62301	0.11288	−5.519	3.41e−08		
Age_group.Q	−0.12860	0.11119	−1.157	0.247450		
HS_Internet.L **	1.39276	0.49218	2.830	0.004658		
HS_Internet.Q	−0.21047	0.43643	−0.482	0.629620		
HS_Internet.C	−0.85687	0.53954	−1.588	0.112255		
HS_Internet ^4	0.59020	0.43162	1.367	0.171499		
Education.L	0.48389	0.36469	1.327	0.184561		
Education.Q	−0.23745	0.34355	−0.691	0.489459		
Education.C	0.22580	0.27517	0.821	0.411902		
Education ^4	−0.22786	0.21790	−1.046	0.295689	2586.074	0.272
Education ^5	0.12788	0.16471	0.776	0.437509		
Education ^6	0.04178	0.15837	0.264	0.791911		
SWB.L ***	0.69000	0.20170	3.421	0.000624		
SWB.Q	0.06799	0.17061	0.399	0.690244		
SWB.C *	0.34709	0.14679	2.365	0.018049		
SWB ^4	−0.14023	0.11854	−1.183	0.236825		
Income.L *	0.34908	0.14307	2.440	0.014688		
Income.Q	0.16223	0.12813	1.266	0.205491		
Income.C	−0.04957	0.12371	−0.401	0.688643		
Q7: Realistic VR/AR scenarios						
Age_group.L ***	−0.69905	0.11221	−6.230	4.67e−10		
Age_group.Q	−0.02388	0.11106	−0.215	0.82974		
HS_Internet.L **	1.40024	0.43716	3.203	0.00136		
HS_Internet.Q	−0.15932	0.39417	−0.404	0.68607		
HS_Internet.C	−0.58856	0.52236	−1.127	0.25985		
HS_Internet ^4	0.25734	0.42615	0.604	0.54593		
Education.L	0.66860	0.35454	1.886	0.05932		
Education.Q	−0.15517	0.33071	−0.469	0.63893		
Education.C	0.20286	0.26793	0.757	0.44896		
Education ^4	−0.22262	0.21357	−1.042	0.29724	2661.548	0.286
Education ^5	−0.04911	0.16301	−0.301	0.76321		
Education ^6	−0.01761	0.15524	−0.113	0.90968		
SWB.L *	0.50940	0.20181	2.524	0.01160		
SWB.Q	0.09199	0.17197	0.535	0.59270		
SWB.C *	0.31064	0.14669	2.118	0.03420		
SWB ^4	−0.08612	0.11836	−0.728	0.46688		
Income.L	0.22028	0.14284	1.542	0.12305		
Income.Q	0.02347	0.12716	0.185	0.85357		
Income.C	0.05636	0.12377	0.455	0.64885		

Table 4. *Cont.*

	Coefficients:				AIC	Nagelkerke's R^2
	Estimate	**Std. Error**	**z Value**	**Pr (>|z|)**		
Q8: Level of interaction						
Age_group.L ***	−1.25567	0.13966	−8.991	<2e−16		
Age_group.Q	0.07930	0.13152	0.603	0.5465		
HS_Internet.L	1.31806	0.87840	1.501	0.1335		
HS_Internet.Q	−0.30481	0.76007	−0.401	0.6884		
HS_Internet.C	−0.39791	0.84723	−0.470	0.6386		
HS_Internet ^4	0.17942	0.63397	0.283	0.7772		
Education.L	0.33534	0.40161	0.835	0.4037		
Education.Q	0.05474	0.38019	0.144	0.8855		
Education.C	0.20631	0.30415	0.678	0.4976		
Education ^4	−0.18008	0.24092	−0.747	0.4548	1463.845	0.329
Education ^5	0.02055	0.18664	0.110	0.9123		
Education ^6	0.13149	0.18232	0.721	0.4708		
SWB.L *	0.57740	0.23667	2.440	0.0147		
SWB.Q	0.39023	0.19911	1.960	0.0500		
SWB.C	0.27184	0.16877	1.611	0.1072		
SWB ^4	−0.07305	0.13992	−0.522	0.6016		
Income.L	0.15605	0.16751	0.932	0.3516		
Income.Q	0.02188	0.15119	0.145	0.8849		
Income.C	−0.15710	0.14518	−1.082	0.2792		
Q9: In-person workshops						
Age_group.L ***	−0.66038	0.10922	−6.047	1.48e−09		
Age_group.Q	−0.02461	0.10897	−0.226	0.82132		
HS_Internet.L **	1.52186	0.49933	3.048	0.00231		
HS_Internet.Q	−0.22160	0.44370	−0.499	0.61747		
HS_Internet.C	−0.61906	0.64554	−0.959	0.33757		
HS_Internet ^4	0.52639	0.51058	1.031	0.30256		
Education.L	0.62519	0.33692	1.856	0.06351		
Education.Q	0.22651	0.31252	0.725	0.46859		
Education.C	0.39415	0.25453	1.549	0.12150		
Education ^4	−0.01394	0.20570	−0.068	0.94596	2771.139	0.299
Education ^5	0.10733	0.15847	0.677	0.49824		
Education ^6	−0.05796	0.15201	−0.381	0.70297		
SWB.L ***	0.88891	0.20351	4.368	1.25e−05		
SWB.Q *	0.43277	0.17096	2.531	0.01136		
SWB.C **	0.36967	0.14238	2.596	0.00942		
SWB ^4	−0.13812	0.11580	−1.193	0.23297		
Income.L	0.21414	0.13916	1.539	0.12385		
Income.Q	−0.01375	0.12429	−0.111	0.91190		
Income.C	−0.01564	0.12057	−0.130	0.89677		
Q10: Gamified elements						
Age_group.L ***	−1.29249	0.12153	−10.635	<2e−16		
Age_group.Q	0.02604	0.11877	0.219	0.82644		
HS_Internet.L **	1.62199	0.53969	3.005	0.00265		
HS_Internet.Q	−0.40984	0.48217	−0.850	0.39533		
HS_Internet.C	−0.64261	0.65791	−0.977	0.32870	2395.537	0.381
HS_Internet ^4	0.40191	0.52363	0.768	0.44276		
Education.L	0.30744	0.37640	0.817	0.41404		
Education.Q	0.59481	0.35319	1.684	0.09216		
Education.C	0.44470	0.28228	1.575	0.11517		

Table 4. *Cont.*

	Coefficients:				AIC	Nagelkerke's R^2
	Estimate	**Std. Error**	**z Value**	**Pr (>|z|)**		
Education ^4 *	0.46301	0.22296	2.077	0.03783		
Education ^5	0.14691	0.16859	0.871	0.38352		
Education ^6	−0.11629	0.15841	−0.734	0.46289		
SWB.L *	0.46347	0.21427	2.163	0.03054		
SWB.Q *	0.37683	0.18057	2.087	0.03689	2395.537	0.381
SWB.C	0.07426	0.15303	0.485	0.62747		
SWB ^4	0.03927	0.12423	0.316	0.75191		
Income.L	−0.02479	0.14948	−0.166	0.86830		
Income.Q	0.12719	0.13518	0.941	0.34678		
Income.C	0.12260	0.12860	0.953	0.34040		
Q11: Level of gamification						
Age_group.L ***	−1.429160	0.143882	−9.933	<2e−16		
Age_group.Q	0.161951	0.140547	1.152	0.24920		
HS_Internet.L	1.581531	0.829155	1.907	0.05647		
HS_Internet.Q	−0.960932	0.733034	−1.311	0.18989		
HS_Internet.C	0.313752	0.735522	0.427	0.66969		
HS_Internet ^4	−0.282181	0.560216	−0.504	0.61447		
Education.L	0.609776	0.417032	1.462	0.14369		
Education.Q	−0.427453	0.389882	−1.096	0.27292		
Education.C **	1.019767	0.315124	3.236	0.00121		
Education ^4	0.047980	0.251492	0.191	0.84870	1351.927	0.348
Education ^5	0.328634	0.193738	1.696	0.08983		
Education ^6	0.096401	0.183517	0.525	0.59937		
SWB.L	0.399468	0.239561	1.668	0.09541		
SWB.Q	0.280259	0.203307	1.379	0.16805		
SWB.C	0.202821	0.173735	1.167	0.24304		
SWB ^4	−0.008207	0.144858	−0.057	0.95482		
Income.L	0.084860	0.172077	0.493	0.62190		
Income.Q	−0.032623	0.155178	−0.210	0.83349		
Income.C	−0.025687	0.148126	−0.173	0.86233		
Q12: Online platform tailored to age groups						
Age_group.L *	−0.24570	0.11115	−2.211	0.02707		
Age_group.Q	0.21378	0.10908	1.960	0.05002		
HS_Internet.L **	1.31077	0.45359	2.890	0.00386		
HS_Internet.Q	−0.24275	0.40691	−0.597	0.55079		
HS_Internet.C	−0.65617	0.51628	−1.271	0.20374		
HS_Internet ^4	0.64289	0.42107	1.527	0.12680		
Education.L	0.23821	0.34069	0.699	0.48442		
Education.Q	0.04149	0.31458	0.132	0.89507		
Education.C	0.27336	0.25714	1.063	0.28775		
Education ^4	0.10899	0.20881	0.522	0.60171	2703.807	0.253
Education ^5	0.18429	0.16159	1.141	0.25408		
Education ^6	−0.12963	0.15186	−0.854	0.39332		
SWB.L **	0.63709	0.20129	3.165	0.00155		
SWB.Q **	0.48165	0.16866	2.856	0.00429		
SWB.C *	0.32047	0.14338	2.235	0.02541		
SWB ^4	−0.09999	0.11697	−0.855	0.39264		
Income.L	0.13409	0.14079	0.952	0.34089		
Income.Q	0.13810	0.12609	1.095	0.27341		
Income.C *	−0.24305	0.12280	−1.979	0.04779		

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; L = linear trend, Q = quadratic trend, C = cubic trend; HS_Internet: high speed internet; and SWB: subjective well-being.

As seen in Table 4, our H1 (*As age increases, preference for online learning mechanisms, in general, will decline*) has been supported, as participants' preferences for most online tools drop steadily with age, except for watching videos of scam scenarios with human actors. Consistent with H1a (*Older age groups will be less intrigued than younger ones to learn through gamified and interactive platforms such as VR/AR*), the steepest age-related declines appear in the most technologically complex formats—especially VR/AR and gamification—where older adults' willingness to engage drops sharply and linearly across the age spectrum, reflecting greater sensitivity to usability demands, effort costs, and high-interactivity interfaces. In sum, the linear age effect is negative and highly significant for most outcomes; for example, gamified elements (Q10):$\beta = -1.292$, $p < 2e{-}16$, indicating a general decline with age. Age is strongly negative for VR-enhanced learning (Q5): $\beta = -0.525$, $p = 1.85e{-}06$, and for AR-enhanced learning (Q6): $\beta = -0.623$, $p = 3.41e{-}08$, supporting H1a.

In contrast, there was no significant ($\beta = -0.073$, $p = 0.503$) difference between younger and older age groups regarding the preference toward watching non-interactive videos with human actors play-acting scam scenarios (Q4), and the model's fit is modest (Nagelkerke $R^2 = 0.260$). Actor-based videos showed no age differences; interest was similar across groups. Thus, H1b (*older age groups will be more interested in learning through less interactive but more real-life stories such as actor-based videos showing real-life case scenarios*) is not supported.

When it comes to further indicators other than age, income level did not show significance with participants' preferences in most online learning tools. Income showed little influence on preferences and rarely predicted learning choices. However, the data show a slight association between income level and some items. The clearest positives are VR (Q5; Income.L $\beta = 0.341$, $p = 0.015$) and AR (Q6; Income.L $\beta = 0.349$, $p = 0.015$), as higher income predicted slightly more (linear) preference towards VR/AR enhanced learning, and slightly less (nonlinear) preference towards creating online platforms tailored to age groups. In the rest of the items, however, income levels did not predict any preferences, regardless of age. Thus, our H2 (*Income level positively correlates with openness to various online methods*) is not supported. In sum, income effects are weak, mostly nonsignificant, and show no strong linear or nonlinear trends across the majority of learning tools, independent of age group. Only VR/AR tools show a clear, small linear increase in interest along with income increase. Overall, income is not a reliable predictor of online learning openness.

Our H3 *(education level will positively correlate with openness to various online methods)* is partially supported. Higher education level indicated a stronger preference toward four items: positive linear and nonlinear effects in interactive features, as the more highly educated the participant, the more preference they had towards interactive features in online learning (Q2). However, this preference unevenly increases with education level (nonlinear effect, curvature increase; Q2 Education.L $\beta = 0.856$, $p = 0.017$ and Q3 Education.L $\beta = 1.128$, $p = 0.001$). Next, higher education levels also indicated an increasing preference towards more interactive elements (Q3) again, with both linear and nonlinear effects being positive and significant. Finally, positive, nonlinear association can be seen between education levels and preferences for gamified elements (Q10) and the level of gamification in the tasks (Q11), although the nonlinear connection suggests that education level alone is not predictive of preferring gamified tasks. In sum, education shows meaningful effects only for certain learning tools: adults with higher education levels express greater interest in interactive features, and this increase is nonlinear—preferences rise more sharply at higher education levels. Thus, education's effects on gamification are uneven and not simply increasing with more schooling. For all other online learning formats, including VR/AR, videos, and workshops, education does not significantly shape preferences.

Our next hypothesis (*H4: Subjective well-being will be associated with greater preferences for online learning methods*) is broadly supported. For interactive elements (Q2), the effect of subjective well-being (SWB) is strongest and positive (SWB.Q $\beta = 0.671$, $p = 0.000113$); beyond the linear term, the nonlinear terms are also significant, indicating a positive relationship that exhibits curvature rather than a simple stepwise increase. For human actor-based scenarios (Q4), the positive nonlinear effect is significant. For VR-enhanced learning (Q5), the linear effect is positive and significant—higher SWB is associated with more favorable evaluations of VR elements in learning—yet the nonlinear component is significant as well. For AR-enhanced learning (Q6), the positive linear effect is likewise significant, with evidence of additional nonlinear patterning. For realistic VR/AR scenarios (Q7), both linear and nonlinear effects are significant. For level of interaction (Q8), the positive linear effect is significant. For in-person workshops (Q9), the linear effect is significant, with nonlinear effects also being observed; however, they are non-significant. For gamification (Q10) and for age-tailored online platforms (Q12), both linear and nonlinear effects are significant. In sum, SWB influences online learning preferences independent of age: the more satisfied individuals are, the more receptive they are to various online learning tools, such as interactive, VR/AR-enhanced, video-based, gamified, and even in-person formats, though in many cases this increase follows a curved rather than strictly linear pattern. Overall, the findings indicate that individuals who feel more satisfied and supported in their daily lives are reliably more willing to engage with a wide range of cybersecurity learning methods.

Finally, H5 *(reliable, high speed internet access will be associated with stronger preference for online learning methods*) has been fully supported. With the exception of a few variables (Q3: frequency of interactive elements, Q8: level of interaction, Q10: gamified elements, and Q11: level of gamification), the effect is significant across all outcomes. As expected, the faster the respondent's internet connection, the greater their receptivity to online learning. Adults with faster connections are consistently more receptive to formats that require stable bandwidth—such as VR/AR tools, realistic scenario videos, and age-tailored platforms—highlighting the practical role of connectivity in enabling participation. Only a few lower-bandwidth outcomes involving simple interactivity showed no effect, underscoring that internet speed mainly shapes preferences for more resource-intensive learning methods.

Overall, the model fit improved significantly after including additional predictors, indicating that while the age group does have a statistically significant effect, its influence is meaningful but moderate (Nagelkerke's R^2 ranges 0.25–0.38, lowest in online platform tailored for age groups, and highest for gamified elements, indicating meaningful but not exhaustive explanatory power once internet access and SWB are included). The most consistently significant variables alongside age group were internet access and subjective well-being, indicating a linear association, suggesting that these factors may exert a greater influence on individuals' likelihood to engage in various online learning methods if better internet access opportunities were available. The results also suggest that as age increases, preferences for all online learning methods decline linearly. By contrast, preferences increase with higher educational attainment and with greater subjective well-being.

5. Discussion

This study investigated the online learning preferences of different age groups in the United States, based on a US adult survey sample that was representative of age. The findings highlight the importance of developing age-specific cybercrime prevention education to enhance user engagement and effectiveness. Survey results reveal distinct

preferences across age groups that inform the development of cybersecurity programs and strategies for target hardening.

5.1. Age-Related Patterns and Theoretical Alignment

Our findings broadly match the theoretical explanations we outlined earlier. First, the robust, negative linear associations between age and most outcomes (e.g., interactivity, VR/AR, gamification) align with andragogy [19], socioemotional selectivity theory (SST; Carstensen et al. [9]), and the technology acceptance model (TAM; Davis [10]) taken together. As andragogy posits, older adults are more selective and purpose-driven; SST predicts a stronger preference for efficient, meaningful engagement as time horizons narrow; and TAM highlights stricter thresholds for *perceived usefulness* and *ease of use*, with rising costs of effort and error. The pattern we observe—lower enthusiasm for online mechanisms as age increases—fits this triad and mirrors prior evidence that motivation translates less into complex performance for older learners under a high cognitive load and unfamiliar interfaces (e.g., [13,16,17]). Hence, these theoretical frameworks should be integrated when designing cybersecurity training for diverse age groups.

The steep age declines for VR/AR and gamified approaches reinforce older age groups' aversions towards technology and the mechanism behind it. These formats add novelty, sensorimotor demands, and rapid decision cycles—precisely the features that reduce *perceived ease of use* and *perceived control* in TAM. These features also conflict with SST's emphasis on efficient goal pursuit. In practical terms, these results caution against deploying highly interactive platforms for older audiences without predictable controls, slower pacing, just-in-time guidance, and clear demonstrations of practical payoff.

5.2. Unexpected Null Effects and Refinements to Experience-Based Learning

Unexpectedly, older adults did not prefer actor-based scenario videos more than younger adults. This null age effect does not undermine the logic of experience-anchored learning in andragogy. However, it does suggest that realistic videos alone are not enough to boost older adults' interest. Two interpretations are plausible. Either (a) actor-based videos are broadly appealing across ages (hence no gradient), or (b) any advantage for older learners depends on additional features we did not manipulate (e.g., tailoring scenarios to personally relevant risks, allowing structured reflection prompts, or pairing videos with checklists). Future experiments should test how adding personal relevance and learner control affects these videos' impact. For practice, this means cybersecurity training designers should add concrete supports—such as scenarios featuring scams that disproportionately target older adults (e.g., Medicare fraud, grandparent scams), short pause-and-reflect questions after each scene, or simple step-by-step "what to do next" checklists—to make narrative videos more actionable for older learners. For theory, it suggests that experience-anchored learning may only produce age-specific benefits when narrative materials are paired with explicit relevance cues and cognitive scaffolds, rather than relying on realism alone to trigger older adults' motivational advantages.

5.3. Socioeconomic Predictors and Structural Enablers

Turning to structural predictors, income did not predict most preferences (limited positives for VR/AR only), whereas education did, most clearly for interactive features and their frequency. This mixed picture comports with the digital-divide framework [25]: income may be a prerequisite for devices and subscriptions, but skills, not just access, shape *preferences*. In other words, education may cultivate the confidence and strategy use that make interactive formats feel worthwhile; income without those skills does less to shift stated preferences.

The subjective well-being (SWB) results were consistently positive and often curvilinear. That is, a higher SWB is associated with stronger preferences for a wide range of modalities, but the relation shows diminishing returns or inflection points. This pattern dovetails with the motivational core of SST: when people feel better off, they can allocate effort to learning that promises meaningful gains; once basic needs are met, additional improvements in SWB may yield smaller marginal increases in preference. The practical implication is to frame online offerings around clear, near-term benefits (e.g., scam avoidance, health self-management) that plausibly enhance SWB, and to build lightweight on-ramps so early successes reinforce motivation.

Finally, the consistently positive effects of reliable high-speed internet support our expectations and underscore a policy lever that is orthogonal to pedagogy: infrastructure matters. Even the best-designed content will underperform where connections are slow or unstable. Coupled with R^2 values that are highest for gamification, these results suggest that access quality and psychosocial readiness (SWB) are at least as consequential as demographic factors for shaping preferences, and that improving connectivity is likely to raise receptivity across formats.

5.4. Implications

Taken together, these results argue for an approach that treats connectivity, skills, and motivation as jointly determinative. Reliable broadband remains a necessary condition, but its benefits materialize only to the extent that older adults are engaged users of information technology. A growing subset already goes online not just to communicate but to pursue cognitive- and knowledge-enhancing activities [8], which positions them to capitalize on richer e-learning opportunities when access is dependable. At the same time, older adults are heterogeneous; many remain at risk of digital exclusion. Platforms that offer flexible pathways—clear, low-friction entry points for novices and deeper, self-directed tracks for more experienced users—are therefore essential [59]. Course design should foreground immediate purpose, predictable interaction, and story-anchored, problem-focused content, while also cultivating the social dimensions of learning. Participation in online communities can increase engagement, motivation, and perceived control among older learners; embedding structured peer exchange and moderated discussion alongside instruction may thus translate access into sustained preference and uptake [59]. Practically, this means that for older learners, VR activities should be paired with stable and minimal controls, reducing complexity and unpredictability in how the user interacts with the virtual environment. Additional elements can include a guided walk-through, and a clearly stated practical goal (e.g., “spot the red flags in this scam scenario”). Designers of narrative or video-based content should also include concrete supports, such as built-in pause points with reflection prompts and one-page “what to do next” checklists, to make learning immediately actionable and to reduce cognitive load.

5.5. Future Work

Future work should experimentally manipulate perceived usefulness, ease of use, and personal relevance within each modality (including community features) to identify the lowest-cost combinations that raise acceptance among older adults with varying skill baselines. Longitudinal designs that track changes in broadband reliability, digital skills training, and subjective well-being can help disentangle age from cohort and test whether improvements in infrastructure and community participation shift preferences over time. In policy terms, investments should pair affordable, reliable broadband with targeted skills support and course designs that acknowledge heterogeneity—so that those who are already engaged can deepen their learning trajectories, while those at risk of exclusion gain on-

ramps that convert access into meaningful educational participation [8,59]. Finally, because our sample includes adults across the full 18+ age range, future research should also examine whether these design principles operate similarly for younger adults or whether explicitly intergenerational learning formats yield additional benefits across age groups.

5.6. Limitations

This study has several limitations that caution the interpretation of its findings. First, the data are cross-sectional, which prevents causal inference and makes it impossible to disentangle age effects from cohort effects. Second, the sample is designed to be representative of age only; other dimensions of population heterogeneity are not guaranteed to be represented, so generalizability beyond age strata is limited. Third, our models necessarily omit potentially important confounders and mechanisms—for example, prior technology experience, digital literacy, cognitive and health status, race/ethnicity, region, household composition, and neighborhood broadband availability—each of which could shape both access and preferences. Relatedly, key variables are self-reported, raising the possibility of measurement error and social-desirability bias. To address these limitations, future research should combine quantitative approaches (e.g., longitudinal panels, field experiments that manipulate perceived usefulness/ease-of-use, and richer controls) with qualitative methods (e.g., interviews and think-aloud usability studies) to uncover the "why" behind preferences—how older adults weigh effort, relevance, and control, and how specific design choices and connectivity conditions influence their decisions over time.

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References

1. Marshall, N.; Sturman, D.; Auton, J.C. Exploring the evidence for email phishing training: A scoping review. *Comput. Sec.* **2024**, *139*, 103695. [CrossRef]
2. Ahmad, N.A.; Abd Rauf, M.F.; Mohd Zaid, N.N.; Zainal, A.; Tengku Shahdan, T.S.; Abdul Razak, F.H. Effectiveness of instructional strategies designed for older adults in learning digital technologies: A systematic literature review. *SN Comput. Sci.* **2022**, *3*, 130. [CrossRef]
3. Internet Crime Complaint Center (IC3). Elder Fraud. Internet Crime Complaint Center. 2024. Available online: https://www.ic3.gov/Outreach/Brochures/elder_fraud_tri-fold.pdf (accessed on 20 August 2025).
4. Federal Trade Commission (FTC). *Protecting Older Consumers 2023–2024. A Report of the Federal Trade Commission*; Federal Trade Commission: Washington, DC, USA, 2024. Available online: https://www.ftc.gov/system/files/ftc_gov/pdf/federal-trade-commission-protecting-older-adults-report_102024.pdf (accessed on 20 August 2025).

5. Neil, L.; Healy, C.; Haney, J. "A five-year-old could understand it" versus "This is way too confusing": Exploring Nonexpert Understandings and Perceptions of Cybersecurity Definitions. *CHI Conf. Hum. Factors Comput. Syst.* **2025**, *5*, 1–9. [CrossRef]
6. Pangrazio, L.; Godhe, A.-L.; González López Ledesma, A. What is digital literacy? A comparative review of publications across three language contexts. *E-Learn. Digit. Media* **2020**, *17*, 442–459. [CrossRef]
7. Kebede, A.S.; Ozolins, L.L.; Holst, H.; Galvin, K. Digital engagement of older adults: Scoping review. *J. Med. Internet Res.* **2022**, *24*, e40192. [CrossRef]
8. van Deursen, A.J.; Helsper, E.J. A nuanced understanding of internet use and non-use among the elderly. *Eur. J. Commun.* **2015**, *30*, 171–187. [CrossRef]
9. Carstensen, L.L.; Isaacowitz, D.M.; Charles, S.T. Taking time seriously. A theory of socioemotional selectivity. *Am. Psychol.* **1999**, *54*, 165–181. [CrossRef] [PubMed]
10. Davis, F.D. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Q.* **1989**, *13*, 319–340. [CrossRef]
11. Hess, T.M. Selective engagement of cognitive resources: Motivational influences on older adults' cognitive functioning. *Perspect. Psychol. Sci.* **2014**, *9*, 388–407. [CrossRef]
12. Tournier, I.; Jordan, O.; Ferring, D. Motivation and memory. *GeroPsych* **2016**, *29*, 147–154. [CrossRef]
13. Gegenfurtner, A.; Vauras, M. Age-Related Differences in the Relation between Motivation to Learn and Transfer of Training in Adult Continuing Education. *Contemp. Educ. Psychol.* **2012**, *37*, 33–46. [CrossRef]
14. Vandenbossche, J.; Coomans, D.; Homblé, K.; Deroost, N. The effect of cognitive aging on implicit sequence learning and dual tasking. *Front. Psychol.* **2014**, *5*, 154. [CrossRef]
15. Hadj-Bouziane, F.; Benatru, I.; Brovelli, A.; Klinger, H.; Thobois, S.; Broussolle, E.; Boussaoud, D.; Meunier, M. Advanced Parkinson's Disease Effect on Goal-Directed and Habitual Processes Involved in Visuomotor Associative Learning. *Front. Hum. Neurosci.* **2013**, *6*, 351. [CrossRef]
16. Caine, K.E.; Nichols, T.A.; Fisk, A.D.; Rogers, W.A.; Meyer, B. Age-related differences in learning incidental, environmental information. *Exp. Aging Res.* **2011**, *37*, 17–45. [CrossRef]
17. Mitzner, T.L.; Fausset, C.B.; Boron, J.B.; Adams, A.E.; Dijkstra, K.; Lee, C.C.; Rogers, W.A.; Fisk, A.D. Older Adults' Training Preferences for Learning to Use Technology. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* **2008**, *52*, 2047–2051. [CrossRef]
18. Carstensen, L.L. The Influence of a Sense of Time on Human Development. *Science* **2006**, *312*, 1913–1915. [CrossRef]
19. Knowles, M.S. *Andragogy in Action: Applying Modern Principles of Adult Education*; Jossey-Bass: San Francisco, CA, USA, 1984.
20. Qin, W. Technology Learning and the Adoption of Telehealth Among Community-Dwelling Older Adults During the COVID-19 Outbreak. *J. Appl. Gerontol.* **2022**, *41*, 1651–1656. [CrossRef] [PubMed]
21. Yamashita, T.; López, E.B.; Soligo, M.; Keene, J.R. Older lifelong learners' motivations for participating in formal volunteer activities in urban communities. *Adult Educ. Q.* **2017**, *67*, 118–135. [CrossRef]
22. Huang, F.; Zhang, M.; Wang, S. Changes in Cognitive Function Among Older Adults: A Latent Profile Transition Analysis. *Arch. Gerontol. Geriatr.* **2019**, *80*, 12–19. [CrossRef] [PubMed]
23. Carstensen, L.L.; Hershfield, H.E. Beyond stereotypes: Using socioemotional selectivity theory to improve messaging to older adults. *Curr. Dir. Psychol. Sci.* **2021**, *30*, 327–334. [CrossRef]
24. Ortega-Dela Cruz, R. Pedagogical practice preferences among generational groups of learners: Towards effective twenty-first Towards effective twenty-first century higher education. *J. Univ. Teach. Learn. Pract.* **2020**, *17*, 5. [CrossRef]
25. Van Dijk, J.A.G.M. Digital divide research, achievements and shortcomings. *Poetics* **2006**, *34*, 221–235. [CrossRef]
26. Choi, N.G.; DiNitto, D.M. Internet use among older adults: Association with health needs, psychological capital, and social capital. *J. Med. Internet Res.* **2013**, *15*, e97. [CrossRef]
27. Miller, L.M.S.; Callegari, R.A.; Abah, T.; Fann, H. Digital literacy training for low-income older adults through undergraduate community-engaged learning: Single-group pretest-posttest study. *JMIR Aging* **2024**, *7*, e51675. [CrossRef]
28. Ellison-Barnes, A.; Moran, A.; Linton, S.; Chaubal, M.; Missler, M.; Evan Pollack, C. Limited technology access among residents of affordable senior housing during the COVID-19 pandemic. *J. Appl. Gerontol.* **2021**, *40*, 958–962. [CrossRef]
29. Umoh, M.E.; Prichett, L.; Boyd, C.M.; Cudjoe, T.K.M. Impact of technology on social isolation: Longitudinal analysis from the national health aging trends study. *J. Am. Geriatr. Soc.* **2023**, *71*, 1117–1123. [CrossRef] [PubMed]
30. Choi, N.G.; DiNitto, D.M.; Marti, C.N.; Choi, B.Y. Telehealth use among older adults during COVID-19: Associations with sociodemographic and health characteristics, technology device ownership, and technology learning. *J. Appl. Gerontol.* **2022**, *41*, 600–609. [CrossRef]
31. Blinka, M.D.; Grieb, S.M.; Lee, T.T.; Hogg, S.; Runge, K.L.; Nogueira, A.; Williams, N.; Prichett, L.; Latkin, C.A.; Gallo, J.J.; et al. 'I make myself get busy': Resilience and social connection among low-income older adults living in subsidized housing. *J. Am. Geriatr. Soc.* **2024**, *72*, 3188–3199. [CrossRef]
32. Kraiwanit, T.; Jangjarat, K.; Srijam, A. Factors determining online activities and technology use among older adults in Thailand. *Pertanika J. Soc. Sci. Humanit.* **2023**, *31*, 803–816. [CrossRef]

33. Chung, J.; Winship, J.; Parsons, P.; Falls, K.; Bleich, M. Stakeholder perspectives on implementation of smart speakers for aging in place in low-income senior housing: A qualitative study. *J. Gerontol. Nurs.* **2022**, *48*, 33–39. [CrossRef] [PubMed]
34. Gallo, H.B.; Marshall, L.W.; Levy-Storms, L.; Wilber, K.H.; Loukaitou-Sideris, A. Voices of experience: What do low-income older adults tell us about mobility, technology, and social participation? *J. Appl. Gerontol.* **2022**, *41*, 571–580. [CrossRef] [PubMed]
35. Guo, Y.; Yeung, C.Y.; Chan, G.C.H.; Chang, Q.; Tsang, H.W.H.; Yip, P.S.F. Mobility based on GPS trajectory data and interviews: A pilot study to understand the differences between lower- and higher-income older adults in Hong Kong. *Int. J. Environ. Res. Public Health* **2022**, *19*, 5536. [CrossRef]
36. Chiu, C.J. Relationship between internet behaviors and social engagement in middle-aged and older adults in Taiwan. *Int. J. Environ. Res. Public Health* **2019**, *16*, 416. [CrossRef]
37. Chen, W.C.; Yang, L.; Wang, X.Y. Internet use, cultural engagement, and multi-dimensional health of older adults: A cross-sectional study in China. *Front. Public Health* **2022**, *10*, 887840. [CrossRef]
38. Liu, W.; Li, W.; Mou, J. Does internet usage make middle-aged and older adults feel healthier? Mediating role of social engagement. *Ind. Manag. Data Syst.* **2024**, *124*, 1–28. [CrossRef]
39. Crouch, E.; Gordon, N.P. Prevalence and factors influencing use of internet and electronic health resources by middle-aged and older adults in a us health plan population: Cross-sectional survey study. *JMIR Aging* **2019**, *2*, e11451. [CrossRef]
40. Bansal, N.; Choudhary, H. Growing old in the digital era: A qualitative study of internet use and outcomes among urban Indian older adults. *Work. Older People* **2024**, *28*, 317–335. [CrossRef]
41. Berner, J.S.; Rennemark, M.; Jogréus, C.; Berglund, J. Factors associated with change in internet usage of Swedish older adults (2004–2010). *Health Inform. J.* **2013**, *19*, 152–162. [CrossRef] [PubMed]
42. Huang, L.; Zhang, T. Perceived social support, psychological capital, and subjective well-being among college students in the context of online learning during the COVID-19 pandemic. *Asia-Pac. Edu. Res.* **2022**, *31*, 563–574. [CrossRef]
43. Lin, Y.; Zhang, P.; Ma, X.; Luo, Q. The influence of community physical exercise on the subjective well-being of older adults: The mediating role of social capital and the moderating role of intergenerational support. *Res. Sq.* **2024**. [CrossRef]
44. Kim, J.; Moon, K.; Lee, J.; Jeong, Y.; Lee, S.; Ko, Y.G. Online learning performance and engagement during the COVID-19 pandemic: Application of the dual-continua model of mental health. *Front. Psychol.* **2022**, *13*, 932777. [CrossRef]
45. Hsu, P.C.; Chang, I.H.; Chen, R.S. Online learning communities and mental health literacy for preschool teachers: The moderating role of enthusiasm for engagement. *Int. J. Environ. Res. Public Health* **2019**, *16*, 4448. [CrossRef]
46. Toledano-González, A.; Labajos-Manzanares, T.; Romero-Ayuso, D. Well-being, self-efficacy, and independence in older adults: A randomized trial of occupational therapy. *Arch. Gerontol. Geriatr.* **2019**, *83*, 277–284. [CrossRef]
47. Adams, K.B.; Leibbrandt, S.; Moon, H. A critical review of the literature on social and leisure activity and wellbeing in later life. *Ageing Soc.* **2011**, *31*, 683–712. [CrossRef]
48. National Telecommunications and Information Administration (NTIA). *Digital Equity Outcomes and Impacts*; National Telecommunications and Information Administration: Washington, DC, USA, 2023. Available online: https://broadbandusa.ntia.gov/sites/default/files/2023-04/Digital_Equity_Outcomes.pdf (accessed on 20 August 2025).
49. Jones, A.; Wetzel, R.; Workman, F. *Bandwidth Requirements for eLearning*; Netforecast Report NFR5141; NetForecast, Inc.: Charlottesville, VA, USA, 2021. Available online: https://www.netforecast.com/wp-content/uploads/NFR5141-eLearning-Bandwidth-Requirements.Final_.pdf (accessed on 20 August 2025).
50. Goldberg, R. *New NTIA Data Show 13 Million More Internet Users in the U.S. in 2023 Than 2021*; National Telecommunications and Information Administration: Washington, DC, USA, 2024. Available online: https://www.ntia.gov/blog/2024/new-ntia-data-show-13-million-more-internet-users-us-2023-2021 (accessed on 20 August 2025).
51. Reichelmann, A.V.; Costello, M. When patriot becomes hate-triot: The relationship between American identity and the production of cyberhate. *Am. J. Crim. Justice* **2021**, *46*, 956–979. [CrossRef]
52. Dearden, T.E.; Parti, K. Cybercrime, differential association, and self-control: Knowledge transmission through online social learning. *Am. J. Crim. Justice* **2021**, *46*, 935–955. [CrossRef]
53. Hawdon, J.; Parti, K.; Dearden, T.E. Cybercrime in America amid COVID-19: The initial results from a natural experiment. *Am. J. Crim. Justice* **2020**, *45*, 546–562. [CrossRef] [PubMed]
54. Kemp, S.; Buil-Gil, D.; Moneva, A.; Miró-Llinares, F.; Díaz-Castaño, N. Empty streets, busy internet: A time-series analysis of cybercrime and fraud trends during COVID-19. *J. Contemp. Crim. Justice* **2021**, *37*, 480–501. [CrossRef]
55. MacInnis, B.; Krosnick, J.A.; Ho, A.S.; Cho, M.-J. The accuracy of measurements with probability and nonprobability survey samples: Replication and extension. *Public Opin. Q.* **2018**, *82*, 707–744. [CrossRef]
56. Simmons, A.D.; Bobo, L.D. Can non-full-probability internet surveys yield useful data? A comparison with full-probability face-to-face surveys in the domain of race and social inequality attitudes. *Sociol. Methodol.* **2015**, *45*, 357–387. [CrossRef]
57. Parti, K.; Dearden, T.; Hawdon, J. Perspectives of paid panel survey research in cybercrime victimization and offending. In *The Routledge International Handbook of Online Deviance*, 1st ed.; Validity of Global Online Market Research Sampling and Data Collection; Graham, R.S., Humer, S.G., Lee, C.S., Nagy, V., Eds.; Routledge: Oxfordshire, UK, 2024; pp. 114–131. [CrossRef]

58. Federal Communications Commission (FFC). *Broadband Data Collection Shows Access to High-Speed Internet Services is Expanding*; Federal Communications Commission: Washington, DC, USA, 2025. Available online: https://www.fcc.gov/news-events/blog/2025/05/20/broadband-data-collection-shows-access-high-speed-internet-services#:~:text=FCC%20Blog-,Broadband%20Data%20Collection%20Shows%20Access%20to,Speed%20Internet%20Services%20is%20Expanding!&text=Since%20June%202022%2C%20the%20FCC,35/3%20Mbps%20or%20greater (accessed on 2 December 2025).
59. Shapira, N.; Barak, A.; Gal, I. Promoting older adults' well-being through internet training and use. *Aging Ment. Health* **2007**, *11*, 477–484. [CrossRef] [PubMed]

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